



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: 高等数学(下) A 开课单位: 数学系
考试时长: 120 分钟 命题教师:

题号	1	2	3	4	5	6	7	8	9
分值	15 分	15 分	10 分	10 分	10 分	10 分	10 分	10 分	10 分

本试卷共 9 道大题, 满分 100 分. (考试结束后请将试卷、答题本、草稿纸一起交给监考老师)

注意: 本试卷里的中文为直译(即完全按英文字面意思直接翻译), 所有数学词汇的定义请参照教材(Thomas' Calculus, 13th Edition)中的定义。如果其中有些数学词汇的定义不同于中文书籍(比方说同济大学的高等数学教材)里的定义, 以教材(Thomas' Calculus, 13th Edition)中的定义为准。

1. (15pts) **Multiple Choice Questions:** (only one correct answer for each of the following questions.)

- (1) If f is differentiable, and $z = z(x, y)$ is determined by $f(x - az, y - bz) = 0$, then $a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} =$
 (A) 1. (B) -1. (C) a (D) b .
- (2) Let $a_n > 0$ for all n . Which of the following statements must be **true**?
 (A) If $\lim_{n \rightarrow \infty} na_n = 0$, then the series $\sum_{n=1}^{\infty} a_n$ converges.
 (B) If $\lim_{n \rightarrow \infty} na_n = l$ and $l \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ converges.
 (C) If $\lim_{n \rightarrow \infty} na_n = l$ and $l \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.
 (D) None of the above statements is correct.
- (3) Identify the surface of $2x^2 + y^2 = z^2$.
 (A) Hyperboloid of two sheets. (B) Elliptical Cone.
 (C) Hyperboloid of one sheet. (D) Elliptical paraboloid.
- (4) If $f(x, y) = \varphi(x + y) + \varphi(x - y) + \int_{x-y}^{x+y} \psi(t)dt$, where φ and ψ are twice differentiable functions, then
 (A) $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial x^2}$. (B) $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y^2}$.
 (C) $\frac{\partial^2 f}{\partial x^2} = -\frac{\partial^2 f}{\partial y^2}$. (D) $\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial y^2}$.
- (5) $\lim_{(x,y) \rightarrow (0,0)} (1 + xy)^{\frac{1}{x^2+y^2}} =$
 (A) 0. (B) 1.
 (C) e . (D) does not exist.

2. (15 pts) Fill in the blanks.

- (1) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are unit vectors and $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$, then $\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a} = \underline{\hspace{2cm}}$.
- (2) If the vector $\mathbf{c}$ is perpendicular to $\mathbf{a} = \langle 1, 2, 1 \rangle$ and $\mathbf{b} = \langle -1, 1, 1 \rangle$ and $\mathbf{c} \cdot (\mathbf{i} - 2\mathbf{j} + \mathbf{k}) = 16$, then $\mathbf{c} = \underline{\hspace{2cm}}$.
- (3) If $\sum_{n=2}^{\infty} \left(\tan \frac{1}{n} - k \ln \left(1 - \frac{1}{n} \right) \right)$ converges, then $k = \underline{\hspace{2cm}}$.
- (4) The maximum curvature κ of function $y(x) = \sin x$ is $\underline{\hspace{2cm}}$.
- (5) If $(z + y)^x = xy$, then $\frac{\partial z}{\partial x}(1, 2) = \underline{\hspace{2cm}}$.

3. (10 pts) Given a cardioid $r = a(1 + \cos \theta)$, $a > 0$ and a circle $r = a$.

- (1) Find the area of the region that lies inside the cardioid and outside the circle.
- (2) Find the area of the region that lies inside the cardioid and inside the circle.

4. (10 pts) Assume $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where

$$f(t) = \int_0^t \cos(x^2) dx, \quad g(t) = -t \cos t, \quad h(t) = \sum_{n=1}^{\infty} \frac{t^n}{n}.$$

Calculate $\mathbf{r}'(0)$.

5. (10 pts) Let $f(x, y) = \begin{cases} y \sin \frac{1}{x^2 + y^2}, & (x, y) \neq (0, 0); \\ 0, & (x, y) = (0, 0). \end{cases}$

- (1) Is $f(x, y)$ is continuous at $(0, 0)$?
- (2) Find $f_x(0, 0)$ and $f_y(0, 0)$, if they exist.

6. (10 pts) Find the limit, if it exists, or show that the limit does not exist.

- (1) $\lim_{(x, y) \rightarrow (0, 0)} \frac{xy}{\sqrt{x^2 + y^2}}.$
- (2) $\lim_{(x, y) \rightarrow (0, 0)} \frac{xy^3 + 2x^2y^4}{x^2 + y^6}.$

7. (10 pts) For the power series $f(x) = \sum_{n=1}^{\infty} \frac{n+2}{n(n+1)} x^n$,

- (1) For what values of x does the power series converge ?
- (2) Find the sum of the series within the interval of convergence.

8. (10 pts) Determine if the series, $\sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n^p (\ln n)^2}$ ($p > 0$), converges absolutely, or converges conditionally, or diverges. Give reasons for your answer.

9. (10 pts) Find $\lim_{n \rightarrow \infty} \left((n^2 - n)e^{\frac{1}{n}} - \sqrt{n^4 + 1} \right).$

一、 (15分) 单项选择题:

- (1) 设二元函数 f 可微, 且 $z = z(x, y)$ 由方程 $f(x - az, y - bz) = 0$ 确定, 则 $a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} =$
 (A) 1. (B) -1. (C) a (D) b .
- (2) 已知对任意 $n > 0$ 都满足 $a_n > 0$, 则下列哪一个结论一定成立?
 (A) 若 $\lim_{n \rightarrow \infty} na_n = 0$, 则级数 $\sum_{n=1}^{\infty} a_n$ 一定收敛.
 (B) 若 $\lim_{n \rightarrow \infty} na_n = l$, 这里 $l \neq 0$, 则级数 $\sum_{n=1}^{\infty} a_n$ 一定收敛.
 (C) 若 $\lim_{n \rightarrow \infty} na_n = l$, 这里 $l \neq 0$, 则级数 $\sum_{n=1}^{\infty} a_n$ 一定发散.
 (D) 以上结论都不对.
- (3) 曲面 $2x^2 + y^2 = z^2$ 是一个
 (A) 双叶双曲面. (B) 椭圆锥.
 (C) 单叶双曲面. (D) 椭圆抛物面.
- (4) 若 $f(x, y) = \varphi(x + y) + \varphi(x - y) + \int_{x-y}^{x+y} \psi(t)dt$, 其中 φ 和 ψ 二阶可导函数, 则
 (A) $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial x^2}$. (B) $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y^2}$.
 (C) $\frac{\partial^2 f}{\partial x^2} = -\frac{\partial^2 f}{\partial y^2}$. (D) $\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial y^2}$.
- (5) $\lim_{(x,y) \rightarrow (0,0)} (1 + xy)^{\frac{1}{x^2+y^2}} =$
 (A) 0. (B) 1.
 (C) e . (D) 极限不存在.

二、 (15分) 填空题:

- (1) 若 $\mathbf{a}, \mathbf{b}, \mathbf{c}$ 为单位向量且满足 $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$, 则 $\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a} =$ _____.
- (2) 向量 $\mathbf{c}$ 垂直于向量 $\mathbf{a} = \langle 1, 2, 1 \rangle$ 和向量 $\mathbf{b} = \langle -1, 1, 1 \rangle$, 且满足 $\mathbf{c} \cdot (\mathbf{i} - 2\mathbf{j} + \mathbf{k}) = 16$, 那么 $\mathbf{c} =$ _____.
- (3) 若 $\sum_{n=2}^{\infty} \left(\tan \frac{1}{n} - k \ln \left(1 - \frac{1}{n} \right) \right)$ 收敛, 则 $k =$ _____.
- (4) 曲线 $y(x) = \sin x$ 的曲率 κ 的最大值为 _____.
- (5) 若 $(z + y)^x = xy$, 则 $\frac{\partial z}{\partial x}(1, 2) =$ _____.

三、 (10分) 设 $a > 0$, 方程 $r = a(1 + \cos \theta)$ 和 $r = a$ 分别表示一条心形线和一个圆周的方程.

- (1) 求心形线所围成的区域中在圆外的部分的面积.
- (2) 求心形线与圆相交的区域面积.

四、 (10分) 设 $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, 其中

$$f(t) = \int_0^t \cos(x^2) dx, \quad g(t) = -t \cos t, \quad h(t) = \sum_{n=1}^{\infty} \frac{t^n}{n}.$$

计算 $\mathbf{r}'(0)$.

五、 (10分) 令 $f(x, y) = \begin{cases} y \sin \frac{1}{x^2+y^2}, & (x, y) \neq (0, 0); \\ 0, & (x, y) = (0, 0). \end{cases}$

(1) $f(x, y)$ 在 $(0, 0)$ 处是否连续?

(2) 若 $f_x(0, 0)$ 和 $f_y(0, 0)$ 存在, 求其值.

六、 (10分) 若下列极限存在, 求其极限值; 若否, 证明其极限不存在.

(1) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2+y^2}}.$

(2) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^3 + 2x^2y^4}{x^2 + y^6}.$

七、 (10分) 设幂级数 $f(x) = \sum_{n=1}^{\infty} \frac{n+2}{n(n+1)} x^n,$

(1) 求幂级数 $f(x)$ 的收敛域.

(2) 求幂级数 $f(x)$ 的和函数.

八、 (10分) 判断级数 $\sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n^p (\ln n)^2} (p > 0)$ 是否为绝对收敛、条件收敛或者发散, 请给出你的理由.

九、 (10分) 求 $\lim_{n \rightarrow \infty} \left((n^2 - n)e^{\frac{1}{n}} - \sqrt{n^4 + 1} \right).$

一、

(1) A

$$f(x-az, y-bz)=0 \Rightarrow \begin{cases} f'_1 \cdot (1-a \frac{\partial z}{\partial x}) + f'_2 \cdot (-b \frac{\partial z}{\partial x}) = 0 \\ f'_1 \cdot (-a \frac{\partial z}{\partial y}) + f'_2 \cdot (1-b \frac{\partial z}{\partial y}) = 0 \end{cases}$$

$$\Rightarrow (1-a \frac{\partial z}{\partial x}) / (1-b \frac{\partial z}{\partial y}) = ab \frac{\partial z}{\partial x} \frac{\partial z}{\partial y}$$

$$a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = 1$$

(2) C

(3) B



(4) D

(5) D

二、

(1) $-\frac{3}{2}$

$$0 = (a+b+c)(a+b+c) = a \cdot a + b \cdot b + c \cdot c + a \cdot b + a \cdot c + b \cdot a + b \cdot c + c \cdot a + c \cdot b$$

$$-2(ab+bc+ca) = 3$$

$$ab+bc+ca = -\frac{3}{2}$$

(2) $c = \frac{8}{5}(1, -3, 3)$

$$\vec{a} = \vec{a} \times \vec{b} = \left(\begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix}, -\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}, \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} \right) = (1, -2, 3)$$

$$\text{let } \vec{c} = \lambda \vec{a}, \quad 16 = \vec{c} \cdot (\vec{i} - 2\vec{j} + \vec{k}) = \lambda(1 + 4 + 3) \Rightarrow \lambda = 2$$

$$\therefore \vec{c} = (2, -4, 6)$$

(3) $k = -1$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\tan \frac{1}{n} - k \ln(1 - \frac{1}{n})}{\frac{1}{n^2}} &= \lim_{x \rightarrow 0} \frac{\tan x - k \ln(1-x)}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{\sin x - k \ln(1-x) \cos x}{x^2 \cos x} \\ &= \lim_{x \rightarrow 0} \frac{1}{\cos x} \cdot \frac{x - \frac{x^3}{3!} + k \left(x + \frac{x^2}{2}\right) \cdot \left(1 - \frac{x^2}{2!}\right) + o(x^2)}{x^2} \\ &= \lim_{x \rightarrow 0} \left(\frac{1+k}{x} + \frac{k}{2} - \left(\frac{1}{3!} + \frac{k}{2!}\right)x - \frac{kx^2}{4} + o(1) \right) \end{aligned}$$

when $k = -1$
 $= -\frac{1}{2}$

(4) $|$

$$\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}$$

$$\begin{cases} x(t) = t \\ y(t) = \sin t \end{cases} \quad \begin{cases} x'(t) = 1 \\ y'(t) = \cos t \end{cases}$$

$$\left| \frac{d\vec{s}}{dt} \right| = |\vec{r}'(t)| = \sqrt{1 + \cos^2 t}, \quad \vec{T} = \frac{\vec{r}'(t)}{|\vec{r}'(t)|} = \frac{1}{\sqrt{1 + \cos^2 t}} (1, \cos t)$$

$$\frac{d\vec{T}}{dt} = \left(\frac{2 \cos t \sin t}{2(1 + \cos^2 t)^{\frac{3}{2}}}, \frac{-\sin t \sqrt{1 + \cos^2 t} + \frac{2 \cos t \sin t \cos t}{2\sqrt{1 + \cos^2 t}}}{1 + \cos^2 t} \right)$$

$$= \left(\frac{\cos t \sin t}{(1 + \cos^2 t)^{\frac{3}{2}}}, \frac{-\sin t}{(1 + \cos^2 t)^{\frac{3}{2}}} \right)$$

$$\left| \frac{d\vec{T}}{dt} \right| = \frac{\sin t \sqrt{1 + \cos^2 t}}{(1 + \cos^2 t)^{\frac{3}{2}}} = \frac{|\sin t|}{1 + \cos^2 t}$$

$$\therefore k(t) = \frac{|d\vec{r}/dt|}{|ds/dt|} = \frac{|\sin t|}{(1+\cos^2 t)^{\frac{3}{2}}} = \sqrt{\frac{\sin^2 t}{(1+\cos^2 t)^3}}$$

$$\min k(t) = k(\frac{\pi}{2} + k\pi) = 1$$

$$(5) \quad 2 - 2\ln 2$$

$$x \ln(z+y) - \ln xy = 0 \quad \text{代入 } \begin{cases} x=1 \\ y=2 \end{cases} \text{ 得 } \ln(z+2) - \ln 2 = 0$$

$$z=0$$

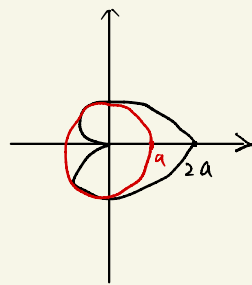
$$\frac{\partial}{\partial x}: \ln(z+y) + \frac{x}{z+y} \frac{\partial z}{\partial x} - \frac{y}{xy} = 0$$

$$\frac{\partial z}{\partial x} = \frac{\frac{1}{x} - \ln(z+y)}{x} (z+y)$$

$$\therefore \frac{\partial z}{\partial x}(1,2) = \frac{\frac{1}{1} - \ln(0+2)}{1} (0+2) = 2 - 2\ln 2$$

$$(1) \quad \frac{a^2(8+\pi)}{4}$$

$$\text{Sol: } \begin{cases} r = a(1+\cos\theta) \\ r = a \end{cases} \Rightarrow \theta = \frac{\pi}{2} + k\pi$$



$$\text{Note: } a(1+\cos\theta) \geq a \quad \text{when } \theta \in [-\frac{\pi}{2} + 2k\pi, \frac{\pi}{2} + 2k\pi]$$

$$a \geq a(1+\cos\theta) \quad \text{when } \theta \in [\frac{\pi}{2} + 2k\pi, \frac{3\pi}{2} + 2k\pi]$$

$$\text{根据图像, calculate } S_{\text{外}} = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} r_{\text{外}}^2 - \frac{1}{2} r_{\text{内}}^2 d\theta$$

$$= \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} a^2(1+\cos\theta)^2 - a^2 d\theta$$

$$\begin{aligned}
&= \frac{a^2}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2 \cos \theta + \cos^2 \theta \, d\theta \\
&= \frac{a^2}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2 \cos \theta + \frac{1 + \cos 2\theta}{2} \, d\theta \\
&= \frac{a^2}{2} \left(2 \sin \theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right) \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\
&= \frac{a^2}{2} \left(4 + \frac{\pi}{2} \right) \\
&= \frac{a^2(8 + \pi)}{4}
\end{aligned}$$

$$(2) \quad \frac{a^2(5\pi - 8)}{4}$$

$$\begin{aligned}
\text{Sol: } S_{r_{\text{in}}} &= 2 \int_0^\pi \frac{1}{2} r_{\text{in}}^2 \, d\theta \\
&= a^2 \int_0^\pi (1 + \cos \theta)^2 \, d\theta \\
&= a^2 \int_0^\pi 1 + 2 \cos \theta + \frac{1 + \cos 2\theta}{2} \, d\theta \\
&= a^2 \left(\frac{3}{2} \theta + 2 \sin \theta + \frac{\sin 2\theta}{4} \right) \Big|_0^\pi \\
&= \frac{3a^2\pi}{2}
\end{aligned}$$

$$\begin{aligned}
\therefore S_{\text{Area}} &= S_{r_{\text{in}}} - S_{\text{out}} \\
&= \frac{3a^2\pi}{2} - \frac{a^2(8 + \pi)}{4} \\
&= \frac{a^2(5\pi - 8)}{4}
\end{aligned}$$

14. $\vec{r}'(0) = \vec{i} - \vec{j} + \vec{k}$

Sol:
$$\begin{cases} f'(t) = \cos(t^2) \\ g'(t) = -\cos t + t \sin t \\ h'(t) = \sum_{n=1}^{\infty} t^{n-1} = \frac{1}{1-t} \end{cases} \Rightarrow \begin{cases} f'(0) = 1 \\ g'(0) = -1 \\ h'(0) = 1 \end{cases}$$

□

五、

(1) $f(x, y)$ 在 $(0, 0)$ 处连续.

Sol: Near $(0, 0)$, $0 \leq |f(x, y)| = |y \sin \frac{1}{x^2 + y^2}| \leq |y|$

$\therefore \lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0 = f(0, 0)$ by sandwich Theorem.

(2) $f_x(0, 0) = 0$, $f_y(0, 0)$ doesn't exist.

Sol: $f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(0+h, 0) - f(0, 0)}{h} = 0$

$$\begin{aligned} f_y(0, 0) &= \lim_{h \rightarrow 0} \frac{f(0, 0+h) - f(0, 0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h \sin \frac{1}{h^2}}{h} \\ &= \lim_{h \rightarrow 0} \sin \frac{1}{h^2} \end{aligned}$$

doesn't exist!

□

六.

(1) 0

Sol: $0 \leq \left| \frac{xy}{\sqrt{x^2+y^2}} \right| = |r \cos \theta \sin \theta| \leq |r|$

$\therefore$ 原式 $= 0$ by sandwich Thm.

(2) 不存在

Sol: along $x = ky^3$

$$\text{原式} = \lim_{(x,y) \rightarrow (0,0)} \frac{ky^6 + 2ky^{10}}{ky^6 + y^6}$$

$$= \frac{k}{1+k}$$

□

七.

(1) $[-1, 1)$

Sol: let $a_n = \frac{n+2}{n(n+1)} = \frac{1}{n} + \frac{1}{n(n+1)}$

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{n+3}{(n+1)(n+2)} \cdot \frac{n(n+1)}{n+2} = 1 \Rightarrow \text{convergence radius } R = 1$$

obvious $\sum a_n = \infty$

$\therefore a_n \searrow 0$

$\therefore \sum a_n(n!)^n < \infty$

$$(2) f(x) = -\frac{\ln(1-x)}{x} - 2\ln(1-x) + 1$$

Sol: $f(x) = \sum_{n=1}^{\infty} a_n x^n = \sum_{n=1}^{\infty} \left(\frac{x^n}{n} + \frac{x^n}{n(n+1)} \right) = g(x) + h(x)$

$$\text{let } \begin{cases} g(x) = \sum_{n=1}^{\infty} \frac{x^n}{n} \\ h(x) = \sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} \end{cases}$$

$$\text{Then } g'(x) = \sum_{n=1}^{\infty} x^{n-1} = \frac{1}{1-x} \Rightarrow g(x) = -\ln(1-x)$$

$$(x h(x))' = \sum_{n=1}^{\infty} x^{n-1} = \frac{1}{1-x}$$

$$\begin{aligned} \Rightarrow x h(x) &= \int -\ln(1-x) dx \\ &= -x \ln(1-x) + \int x d \ln(1-x) \\ &= -x \ln(1-x) - \int \frac{x}{1-x} dx \\ &= -x \ln(1-x) - \int -1 + \frac{1}{1-x} dx \\ &= -x \ln(1-x) + x + \ln(1-x) \end{aligned}$$

$$\Rightarrow h(x) = \frac{\ln(1-x)}{x} - \ln(1-x) + 1$$

$$\therefore f(x) = g(x) + h(x) = \frac{\ln(1-x)}{x} - 2\ln(1-x) + 1$$

D

1、 $p \geq 1$ 绝对收敛 $\leftarrow$ 积分判别法.

$0 < p < 1$ 条件收敛 $\leftarrow$ 交错级数 + 与 $\frac{1}{n}$ 比较判别
洛必达 2 次

9. $-\frac{1}{2}$

Sol: $\lim_{n \rightarrow \infty} [(n^2 - n)e^{\frac{1}{n}} - \sqrt{n^4 + 1}]$

$$= \lim_{n \rightarrow \infty} [(n^2 - n)(1 + \frac{1}{n} + \frac{1}{2!n^2}) - \sqrt{n^4 + 1}]$$

$$= \lim_{n \rightarrow \infty} [n^2 - n + n - 1 + \frac{1}{2} - \frac{1}{2n} - \sqrt{n^4 + 1}]$$

$$= -\frac{1}{2} + \lim_{n \rightarrow \infty} [n^2 - \sqrt{n^4 + 1}]$$

$$= -\frac{1}{2} + \lim_{n \rightarrow \infty} \frac{-1}{n^2 + \sqrt{n^4 + 1}}$$

$$= -\frac{1}{2}$$

□